A MONOMORPHISM DOMINATED BY AN FPPF POINT IS EMPTY OR AN ISOMORPHISM

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Editorial note. Counted span is the author's own proof environment, source lines 3192-3207, immediately following the statement at 3183-3190.

Editorial note. Two cross-chapter pointers (Morphisms of Spaces, Properties of Spaces) and one cross-chapter Remark (Spaces) are recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Fibres of monomorphisms towards a field. Morphisms of Spaces, Lemma spaces-morphisms-lemma-monomorphism-toward-field, cited at proof lines 3197-3199. T = Z' x_Z Spec(k) -> Spec(k) is a monomorphism with nonempty source, hence an isomorphism.
- Nonemptiness of the fibred product. Properties of Spaces, Lemma spaces-properties-lemma-points-cartesian, cited at proof lines 3194-3196. Z' nonempty and Spec(k) -> Z surjective give T nonempty.
- Surjectivity as a map of fppf sheaves. Spaces, Remark spaces-remark-warning, cited at proof lines 3203-3205. A flat, locally of finite presentation, surjective morphism is surjective as a map of sheaves on (Sch/S)_fppf – the pivot for concluding Z' = Z.

Source reference. Original reference 06MG: focused-seeds/geometry/S9-decent-monomorphism-into-point/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03H4: focused-seeds/geometry/S9-decent-monomorphism-into-point/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02YN: focused-seeds/geometry/S9-decent-monomorphism-into-point/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let $Z' \to Z$ be a monomorphism of algebraic spaces over S. Assume there exists a field k and a locally finitely presented, surjective, flat morphism* $\mathrm{Spec}(k) \to Z$. *Then either Z' is empty or $Z' = Z$.*

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Proof. We may assume that Z' is nonempty. In this case the fibre product $T = Z' \times_Z \mathrm{Spec}(k)$ is nonempty, see Properties of Spaces, Lemma 03H4. Now T is an algebraic space and the projection $T \to \mathrm{Spec}(k)$ is a monomorphism. Hence $T = \mathrm{Spec}(k)$, see Morphisms of Spaces, Lemma 06MG. We conclude that $\mathrm{Spec}(k) \to Z$ factors through Z'. But as $\mathrm{Spec}(k) \to Z$ is surjective, flat and locally of finite presentation, we see that $\mathrm{Spec}(k) \to Z$ is surjective as a map of sheaves on $(Sch/S)_{fppf}$ (see Spaces, Remark 02YN) and we conclude that $Z' = Z$. $\square$